Giovanni Tauro

R&D Engineer, PhD Candidate| Wearable Sensor Data & IoT



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LinkedIn GitHub Scholar Portfolio GrabCAD

R&D Engineer and PhD Candidate with 5+ years at CNR-STIIMA building wearable and IoT sensing systems for health research. For OmniTrack I own the data path end to end: UWB and IMU time series from wearable tags, session logging in MongoDB, and a Python pipeline (pandas, FastAPI) that turns study recordings into validated, analysis-ready datasets.

Work Experience

R&D Product Engineer, STIIMA, Lecco
Feb 2021 – Present

- Built wearable and IoT sensing systems for rehabilitation research, collecting time-series data (position, IMU) in studies with patients and clinicians.
- Defined the architecture of multimodal IoT platforms, integrating wearable sensing, indoor localization, BLE communication, and data logging.
- Designed evaluation protocols and analysed usability, system performance, and user interaction data in Python.
- Defined hardware/software requirements with clinicians and deployed PhiCube devices in Italian clinics within Fit4MedRob (MUR, national programme).

`Python` • `pandas` • `NumPy` • `MongoDB` • `ROS2` • `ESP32` • `BLE GATT` • `C++` • `Git` • `Linux`

Head of Product (Co-Founder), Rehabilia Technologies s.r.l., Milan
May 2023 – Present

- Lead product development and prioritisation with engineering, design, and clinical stakeholders, from user research to validation.
- Validate product decisions with usability testing and study data: PhiCube reached SUS 85.8/100 with 19 children and is in two ongoing multicentric studies with 134 patients.

Selected Projects

GIT_1
GIT_2
GIT_3

OmniTrack: IoT-based Multimodal Navigation System /Frontiers

- Built the data pipeline end to end: UWB ranging and IMU time series from wearable ESP32 tags (micro-ROS), position estimation in ROS 2, and session logging in MongoDB.
- Developed OmniAnalytics, a Python/FastAPI backend (pandas, NumPy) with a React dashboard that turns raw session recordings from 16 blind participants into trial metrics: route similarity (DTW, Fréchet distance), speed, and target discovery.
- Kept data traceable: corrections, exclusions, and annotations are stored separately, so raw recordings are never modified.

`Python` • `pandas` • `FastAPI` • `MongoDB` • `React` • `ROS2` • `ESP32` • `UWB` • `IMU`

WEB

PhiCube: reconfigurable bilateral robotic device for neurorehabilitation /Frontiers

- Designed a modular rehabilitation device combining UX research, mechanical design, embedded systems, and multimodal interaction.
- Validated with 19 children (SUS 85.8/100); 134 patients in two on-going multicentric studies.

`UX • Interaction Design • CAD • Prototyping • Assembly • Usability`

GIT

RehaBalloon: Interactive Platform

- Prototyped a low-cost rehabilitation device in 24 hours during the LMU Munich "Hacking with Hardware" challenge, integrating pressure sensing, visual feedback, and embedded electronics.
- Implemented a TensorFlow Lite model on Arduino embedded machine learning classifier exercise patterns and record rehabilitation activity metrics in real time.

`Seeed Studio • Sensors • C++ • TensorFlow • Machine Learning`

Education

2023–27 **PhD, Mechanics and Advanced Engineering Sciences** *Bologna University, IT*

PhD Candidate, expected 2027. Topic: IoT-based Multimodal Navigation System – OmniTrack

- Research Mobility, Utrecht University, NLD: Sensory navigation and experimental psychology.
- Research Mobility, Toyo University, JP: Feedback algorithm and behaviour analysis.

2017–20 **MSc, Design & Engineering** *Politecnico di Milano, IT*

Topic: UX development of an ultrasonic dental device: from concept to validation: WashB

- Research Mobility, Chiba University, JP: UX research studies for Fujitsu and Panasonic.

2012–15 **BSc, Industrial Design** *Florence University, IT*

Topics: Foundation in research methodologies, development, and critical thinking.

Certifications

2026 **Claude 101 Badge — Claude Code Badge — Claude** *Anthropic, Online*

2024 **.ppt — Robotics and Allied Technologies for Rehab** *MUR, Rome, Italy*

Awards

2022–23 **EIT Health MedTech Bootcamp** and **EIT Wildcards Bootcamp**: selected by the European support programmes for innovative medical startups.

Patent

2023 Malosio, M., Nicora, M. L., Redaelli, D., Tauro, G. WO2023214349A1 *Device for bilateral rehabilitation*.

Languages

Italian: Native English: Advanced (IELTS 2017) French: Basic (A2)

Lecco, October 12, 2026



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